

Statistics Method II Assignment

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Question one:

Codes:

```
#Data Importation
mydata <- read.csv("~/R for Statistics and Data Science (STAT-5521G)/Stat_meth.txt", sep="")

#Linear Regression Model
lm_model<-lm(Minutes~Copiers, data = mydata)

#Summary of the LM result
summary(lm_model) -> summary_model
summary_model
```

```
##
## Call:
## lm(formula = Minutes ~ Copiers, data = mydata)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -22.7723  -3.7371   0.3334   6.3334  15.4039
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -0.5802     2.8039  -0.207   0.837
## Copiers       15.0352     0.4831  31.123 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.914 on 43 degrees of freedom
## Multiple R-squared:  0.9575, Adjusted R-squared:  0.9565
## F-statistic: 968.7 on 1 and 43 DF,  p-value: < 2.2e-16
```

(a.) What is the dependent variable in this case;

The dependent variable is: **total number of minutes.**

(b.) What is the independent variable in this case;

The independent variable is: **number of copiers serviced.**

(c.) Show what is the theoretical simple linear regression model;

$$Y = \beta_0 + \beta_1 X + \epsilon$$

where:

- Y = Total service time (minutes) (**dependent variable**),
- X = Number of copiers serviced (**independent variable**),
- β_0 = Intercept (baseline service time when no copiers are serviced),
- β_1 = Slope (additional time required per copier),
- $\epsilon \sim N(0, \sigma^2)$ is the error term (assumed to follow a normal distribution with mean zero and constant variance).

(d.) Can you provide a least squares estimator for the parameter value β_1 ;

$$\beta_1 = 15.0352$$

(e.) Can you provide a least squares estimator for the parameter value;

$$\beta_0 = -0.5802$$

(f.) Show what is the fitted linear regression model;

Total service time (minutes) = $-0.5801567 + 15.035248$ (Number of copiers serviced)

(g.) Can you provide an interpretation for the slope β_1 ;

For one unit increase in number of copiers serviced, will yield to 15.0352 increase in the total serviced time on average.

(h.) Can you provide an interpretation for the intercept β_0 ;

On average the total serviced time is -0.5802 whenever the number of copiers serviced is 0.